

- 8 (a) wave theory predicts any frequency would give rise to emission of electron M1
 if exposure time is sufficiently long A1
 photon has (specific value of) energy dependent on frequency M1
 emission if energy greater than threshold / work function / energy to remove
 electron from surface A1 [4]
- (b) photon is packet/quantum of energy M1
 of electromagnetic radiation A1
 (photon) energy = $h \times \text{frequency}$ B1 [3]
- every particle has an (associated) wavelength B1
 wavelength = h / p M1
 where p is the momentum (of the particle) A1 [3]
- 9 (a) (i) $\Delta N / \Delta t$ (ignore any sign) B1 [1]